From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST-III

PAPER -2

ANSWERS KEY			
	PHYSICS	CHEMISTRY	MATHEMATICS
Q. No.	ANSWER	ANSWER	ANSWER
1.	B	A	A
2.	B	C	D
3.	A	A	D
4.	D	C	C
5.	B	C	C
6.	A	A	D
7.	A	B	B
8.	A	B	D
9.	A, B, C	A, B, D	A, B, C
10.	A, B, C, D	A, C	A, B, C, D
11.	A, B, C	B, C, D	B, D
12.	A, B, C, D	C, D	A, C, D
1.	(A) → (p) (B) → (q) (C) → (q) (D) → (s)	(A) → (r) (B) → (s) (C) → (p) (D) → (q)	(A) → (r) (B) → (p, s) (C) → (p, q, r, s) (D) → (p)
2.	(A) → (q) (B) → (p) (C) → (r) (D) → (s)	(A) → (s) (B) → (r) (C) → (t) (D) → (p)	(A) → (s) (B) → (r) (C) → (q) (D) → (p)
1.	2	6	2
2.	2	3	3
3.	1	4	3
4.	2	4	2
5.	0	9	7
6.	1	3	2

Physics

PART – I

SECTION – A

2. Potential energy will be finite and maximum at x = 0.

5. $B_1 - B_2 = 20\ \mu T$, $B_1 + B_2 = 50\ \mu T$ and $B \propto i$

6. Use $\tau = \vec{M} \times \vec{B}$

7. Use symmetry arguments. Net $\vec{B}$ will be perpendicular to the plane of the loop

9. Use Kirchoff's law and concept of symmetry as well.

10. $\frac{v}{u} = 3$

$\frac{1}{v} - \frac{1}{u} = 1/f$

f = 30 cm

11. $\sin\beta/2 = \sin\alpha/2\sqrt{\frac{q_1}{q_2}}$

SECTION –C

1. $R_{eq} = 1\Omega$

$I_{max} = \frac{4v}{1+1} = 2\ Amp$

2. $E = \frac{1}{2}CV^2 = \frac{1}{2}\alpha\beta t^3$

$\frac{dE}{dt} \alpha\ t^2$

3. Torque = NABI sin θ
Max Torque = NABI for $\theta = 90°$
For a coil of given length L
Radius = $L/2\pi N$

Area = π(radius) = $\pi \times \frac{L^2}{4\pi^2N^2}$

Torque = $NBI\frac{\pi L^2}{4\pi^2N^2}$

Torque $\alpha\ \frac{I}{N}$

$\therefore$ Smallest value of N for max Torque is n = 1

4. $E = \frac{1}{2}CV^2 = \frac{1}{2}\alpha\beta t^3$

$\frac{dE}{dt} \alpha\ t^2$

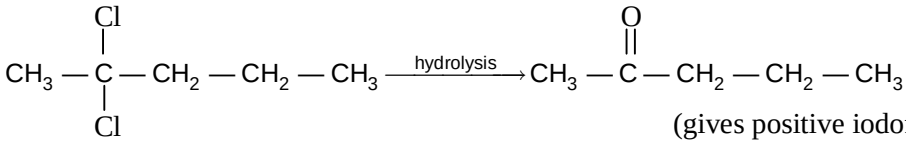
5. The two loops produce equal and opposite magnetic fields.

6. Capacitor is directly connected with 1 V cell in parallel to it
$\therefore$ Q = CV = 1 × 1 = 1 μC.

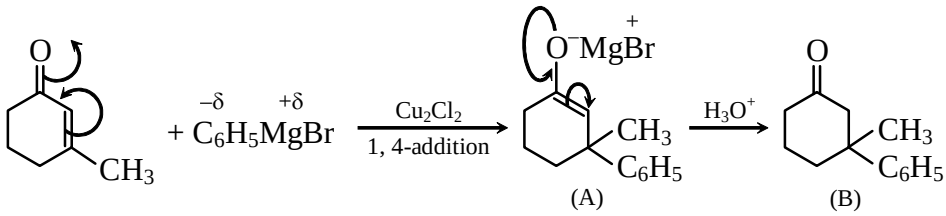
Chemistry

PART – II

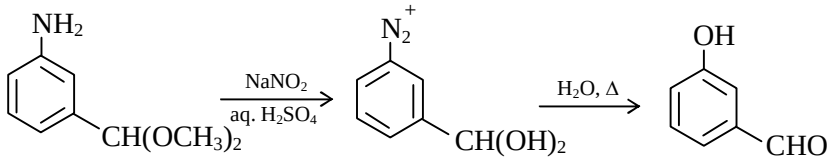
SECTION – A

1. 

(gives positive iodoform test but does not give Fehling's test)

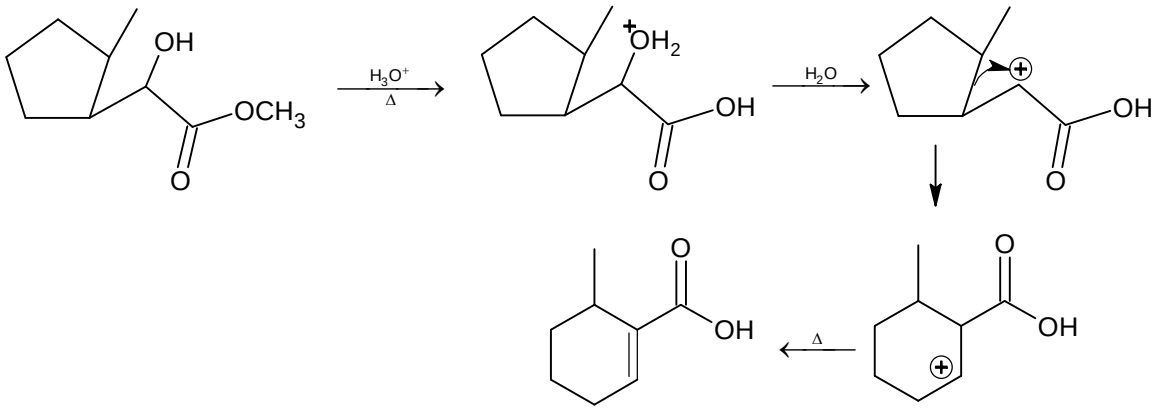
2. 

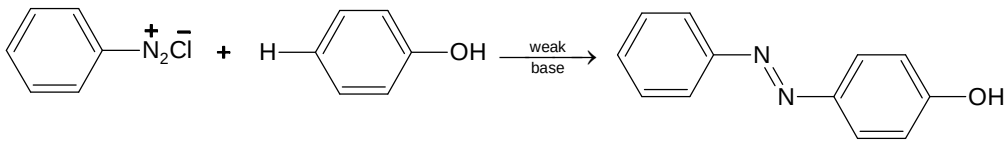
3. Due to neighbouring group participation of OEt group.

4. 

5. $2CuO + C \xrightarrow{\Delta} 2Cu + CO_2\uparrow$

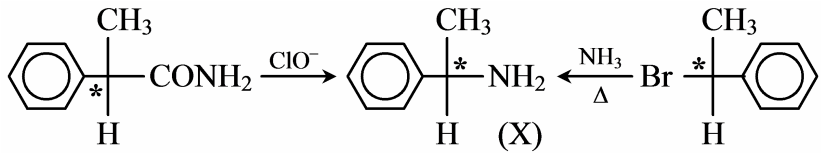
6. In polystyrene the monomeric unit is styrene which is having the formula of $H_2C{=}CH{—}C_6H_5$.

7. 

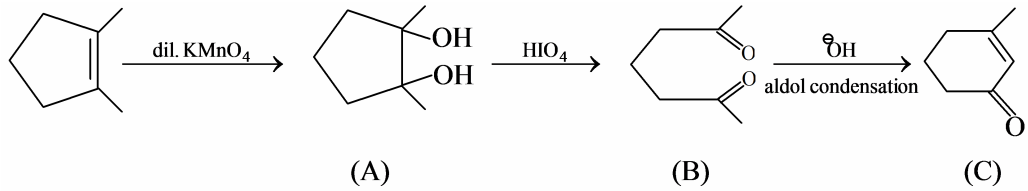
8. 

9. Galactose is empimer of D-(+)-glucose. It is monomer.

11.

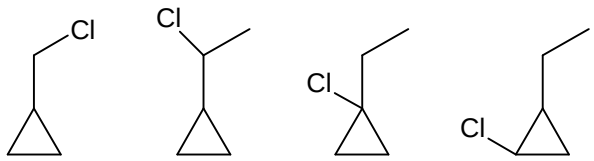


12.



SECTION – C

1.

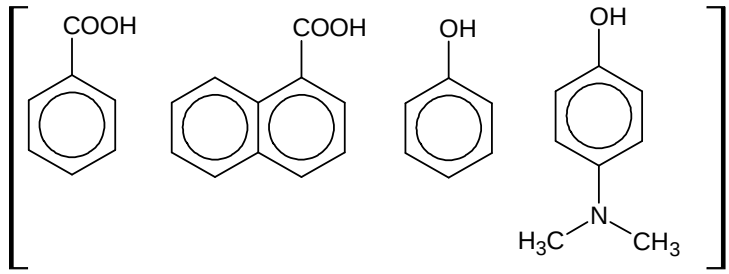


(one) (two) (one) (two)

2. One with keto and two with ester group (lactone).

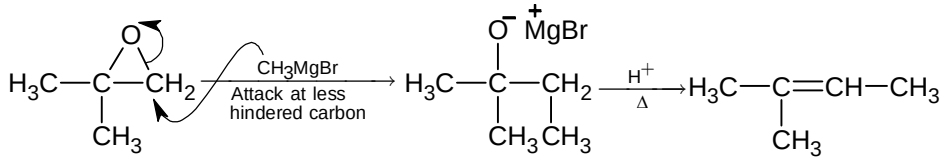
3. All are stereoisomers.

4.

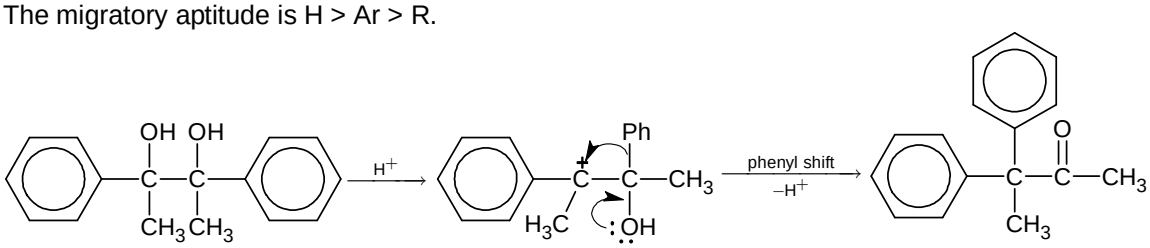


are soluble in alkalies.

5.



6. Pinacol-Pinacolone rearrangement reaction.
The migratory aptitude is H > Ar > R.



Mathematics

PART – III

SECTION – A

1. $f'(x) = -\frac{1}{2}(a^2 - 3a + 2)\sin\frac{x}{2} + (a-1) \neq 0$

$\Rightarrow (a-1)\left(1 - \frac{(a-2)\sin x/2}{2}\right) \neq 0$

$\Rightarrow a \in (0, 1) \cup (1, 4)$

2. $f'(x) = \frac{-2x}{x^2+x+1} + b^2 - 5b + 4 < 0$

$\Rightarrow b^2 - 5b + 4 < \frac{2x}{x^2+x+1}$

$\Rightarrow b^2 - 5b + 4 < -2 \Rightarrow b \in (2, 3)$

3. $PR \leq PQ$

$\Rightarrow (h-a)^2 + k^2 = (a-k)^2$

$\Rightarrow y \leq x - \frac{x^2}{2a}$

$\Rightarrow$ required area $= \int_0^{2a}\left(x - \frac{x^2}{2a}\right)dx = \frac{2a^3}{3}$

4. $y^2 = a^2 - b^2\cos^2(\ln x) \Rightarrow 2y\frac{dy}{dx} = 2b^2\cos(\ln x)\sin(\ln x).\frac{1}{x}$

$\therefore\ x^2y^2\left(\frac{dy}{dx}\right)^2 = b^4\cos^2(\ln x)\sin^2(\ln x) = b^4\left(\frac{a^2-y^2}{b^2}\right)\left(1 - \frac{a^2-y^2}{b^2}\right)$

5. Using approximations

$L = \lim_{x\to0}\frac{\frac{\sin^2 x}{3} + \frac{2\tan x}{4}}{\sin x + \tan^2 x} = \lim_{x\to0}\frac{\frac{x^2}{3} + \frac{2x}{4}}{x + x^2} = \lim_{x\to0}\frac{x\left(\frac{x}{3} + \frac{1}{2}\right)}{x(1+x)} = \frac{1}{2}$

6. $f'(x + 2k) = f'(x)$, hence $f'(x)$ is periodic with period as $2k$

$\Rightarrow \int_a^{2nk+a}\left|\sin\left(\frac{\pi x}{2k}\right)\right| f'(x)\,dx = \int_0^{2nk}\left|\sin\left(\frac{\pi x}{2k}\right)\right| f'(x)\,dx$

$= n\int_0^{2k}\left|\sin\left(\frac{\pi x}{2k}\right)\right| f'(x)\,dx = \frac{-n\pi}{2k}\int_0^{2k}\cos\left(\frac{\pi x}{2k}\right) f(x)\,dx$

7. $f(x) = \cos^{-1}(\sec(\cos^{-1}x)) + \sin^{-1}(\operatorname{cosec}(\sin^{-1}x))$

$\Rightarrow -1 \leq \sec(\cos^{-1}x) \leq 1$ and $-1 \leq \operatorname{cosec}(\sin^{-1}x) \leq 1$

$\Rightarrow \sec(\cos^{-1}x) = \pm 1$ and $\operatorname{cosec}(\sin^{-1}x) = \pm 1$

$\Rightarrow \cos^{-1}x = 0, \pi$ and $\sin^{-1}x = \frac{\pi}{2}, -\frac{\pi}{2}$

$\Rightarrow x = \pm 1$ and $x = \pm 1 \Rightarrow$ domain is $x = \pm 1$

8. Put $x = y = 1 \Rightarrow f(1) = 2$

Put $y = \frac{1}{x} \Rightarrow f(x) + f\left(\frac{1}{x}\right) = f(x)\ f\left(\frac{1}{x}\right)$

$\Rightarrow f(x) = x^3 + 1$

$\Rightarrow g(x) = 0 \ \forall \ x \in R - \{0\}$

9. If we simplify, we get $f(x) = 3\left(x + \frac{1}{x}\right)$.

11. $f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\ln t}{(1+t)} dt = \int_1^x \frac{-\ln u}{\frac{1+u}{u}} \times \frac{-1}{u^2} du = \int_1^x \frac{\ln t}{t(1+t)} dt$

Now, $f(x) + f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln t}{(1+t)}\left(1+\frac{1}{t}\right) dt = \int_1^x \frac{\ln t}{t} dt = \left[\frac{(\ln t)^2}{2}\right]_1^x = \frac{(\ln x)^2}{2}$

12. $\frac{dy}{dx} = \frac{-x^2 - 2x + 1}{(x^2+1)^2}$ and $\frac{d^2y}{dx^2} = \frac{2x^3 + 6x^2 - 6x - 2}{(x^2+1)^3}$

$\frac{d^2y}{dx^2}$ is 0 at three points which are the roots of the equation $x^3 + 3x^2 - 3x - 1 = 0$

i.e. $-2 - \sqrt{3}$, $-2 + \sqrt{3}$, 1.

Checking the sign of $\frac{d^2y}{dx^2}$ in the neighbourhood of these points, all these are the point of inflection.

SECTION – B

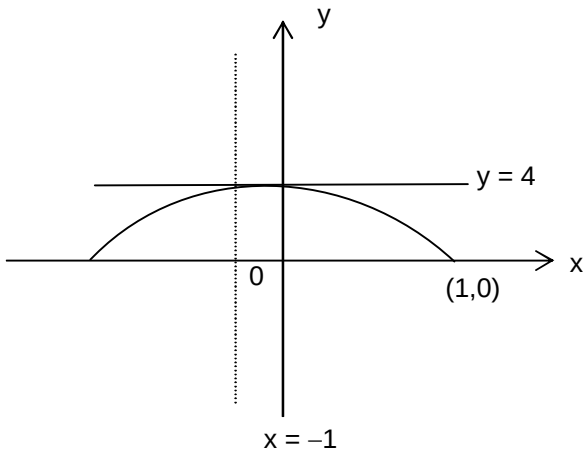
2. (A). Required area = $\int_0^1 \left(\sqrt{x} - x^2\right) dx = 1/3$

(B). $\int_0^4 \left\{\sqrt{x}\right\} dx = \int_0^4 \left(\sqrt{x} - \left[\sqrt{x}\right]\right) dx = \int_0^4 \sqrt{x}\, dx - \int_0^1 \left[\sqrt{x}\right] dx - \int_1^4 \left[\sqrt{x}\right] dx = 7/3$

(C). Area = $\int_0^1 \left(3 - 2x - x^2\right) dx$

$= \left[3x - x^2 - \frac{x^3}{3}\right]_0^1$

$= 5/3$



(D). $2\int_0^{\pi/2} \sqrt{\cos x}\ \sin x\, dx = 4/3$

SECTION–C

1. $t \ln t - t\big|_0^x + x = \frac{x^2}{3} \Rightarrow \ln x = \frac{x}{3}$.

Now equation of tangent of the slope $\frac{1}{3}$ to the curve $y = \ln x$ at $(3, \ln 3)$ is

$y - \ln 3 = \frac{1}{3}(x - 3) \Rightarrow y = \frac{1}{3}x + (\ln 3 - 1)$.

As, $\ln 3 - 1 > 0$, this tangent is above the line $y = \frac{x}{3}$.

So the line $y = \frac{x}{3}$ will cut $y = \ln x$ at two distinct points.

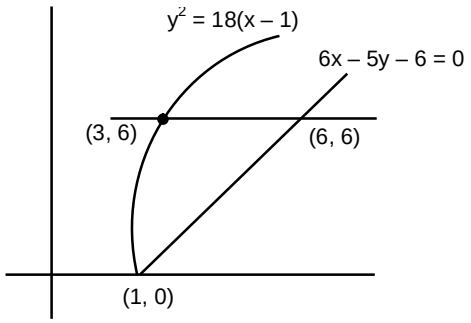
2. $4x^2 + 4x + 4 + \sin(\pi x) = (2x + 1)^2 + \sin(\pi x) + 3 \geq 2$.
Now as $1 < 2 < e$, the required value of n is 3.

3. The function is discontinuous at $x = 0, \frac{1}{2}$ and 2.

4. If $x > 1$
$y = 6$ …(1)
$y^2 = 18(x - 1)$ …(2)
(1) and (2) intersect at (3, 6).
Intersection of $y^2 - 18x + 18 = 0$ and $6x - 5y - 6 = 0$ is (6, 6)

$$A = \int_1^3 \left\{\sqrt{18(x-1)} - \frac{6}{5}(x-1)\right\}dx + \int_3^6 \left\{6 - \frac{6}{5}(x-1)\right\}dx$$

$= 11$.



5. $[\tan x]^2 + \tan x - a = 0$
$\because [\tan x]^2$ and a are integral in nature $\Rightarrow \tan x$ is integer.
$\Rightarrow$ equation reduces to $\tan^2 x + \tan x - a = 0$
$\Rightarrow \tan x (\tan x + 1) = a$
$\therefore$ a is product of consecutive integer
Here a = 12, 20, 30, 42, 56, 72, 90
$\Rightarrow$ a has 7 values.

6. Let $\cos\theta = \frac{\sqrt{1+x^4}}{1+x^2}$, we get

$$-\sin\theta\frac{d\theta}{dx} = \frac{2x^3}{(1+x^2)\left(1+x^4\right)} - \frac{2x\sqrt{1+x^4}}{(1+x^2)^2} = \frac{2x^3(1+x^2) - 2x(1+x^4)}{(1+x^2)^2\sqrt{1+x^4}}$$

$$= \frac{-2x(1-x^2)}{(1+x^2)^2\sqrt{1+x^4}} = \frac{1-x^2}{(1+x^2)\sqrt{1+x^4}}dx = \left(\frac{1+x^2}{2x}\right)\sin\theta\, d\theta$$

The factor on the right can be put in terms of θ as follows

$$\frac{1+x^2}{2x} = \frac{1+x^4+2x^2}{4x^2} = \left(\frac{\sqrt{1+x^4}}{2x}\right)^2 + \frac{1}{2} = \left(\left(\frac{1+x^2}{2x}\right)\cos\theta\right)^2 + \frac{1}{2}$$

$$\Rightarrow \left(1-\cos^2\theta\right)\left(\frac{1+x^2}{2x}\right)^2 = \frac{1}{2} \Rightarrow \frac{1+x^2}{2x} = \frac{1}{\sqrt{2}\sin\theta}$$

$\therefore$ the given integral becomes

$$\frac{8\sqrt{2}}{\pi}\int_0^{\pi/4}\frac{\sin\theta\, d\theta}{\sqrt{2}\sin\theta} = \frac{8\sqrt{2}}{\pi}\cdot\frac{1}{\sqrt{2}}\int_0^{\pi/4}d\theta = \frac{8\sqrt{2}}{\pi}\cdot\frac{1}{\sqrt{2}}\frac{\pi}{4} = 2$$